In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import urllib.request
import zipfile
import io
```

In [3]:
```
url = "https://archive.ics.uci.edu/static/public/222/bank+marketing.zip"
response = urllib.request.urlopen(url)
with zipfile.ZipFile(io.BytesIO(response.read())) as archive:
    csv_path = next(name for name in archive.namelist() if name.lower().e
    with archive.open(csv_path) as nested_file:
        with zipfile.ZipFile(io.BytesIO(nested_file.read())) as bank_arch
            csv_path = next(name for name in bank_archive.namelist() if n
            with bank_archive.open(csv_path) as file:
                df = pd.read_csv(file, sep=";")

X = df.drop(columns=["y"])
X = pd.get_dummies(X, drop_first=True, dtype=float)
X = X.replace([np.inf, -np.inf], np.nan).fillna(0)
X.head()
```

Out[3]:

	age	balance	day	duration	campaign	pdays	previous	job_blue-collar	job_entrepr
0	58	2143	5	261	1	-1	0	0.0	
1	44	29	5	151	1	-1	0	0.0	
2	33	2	5	76	1	-1	0	0.0	
3	47	1506	5	92	1	-1	0	1.0	
4	33	1	5	198	1	-1	0	0.0	

5 rows × 42 columns

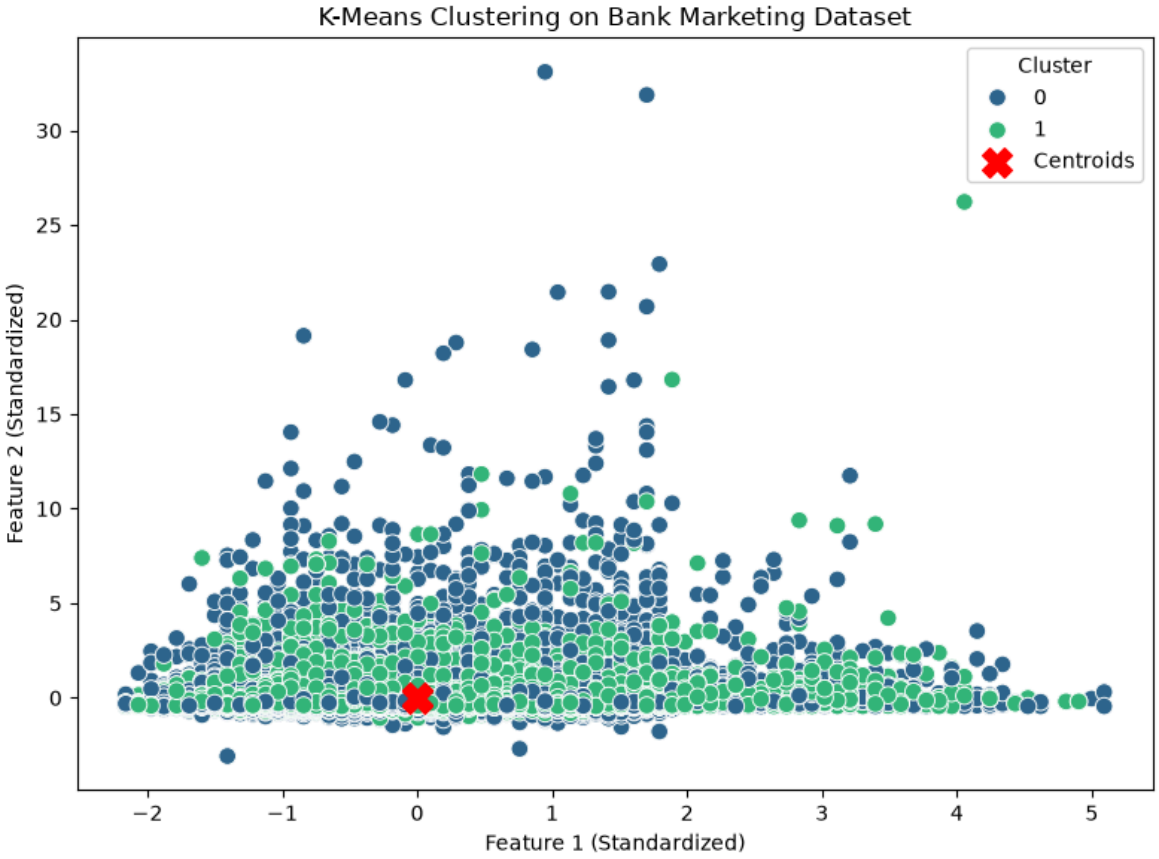
In [4]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

In [5]:
```
kmeans = KMeans(
    n_clusters=2,
    random_state=42,
    n_init=10
)

y_kmeans = kmeans.fit_predict(X_scaled)
```

In [6]:
```
plt.figure(figsize=(8, 6))
sns.scatterplot(
    x=X_scaled[:, 0],
    y=X_scaled[:, 1],
    hue=y_kmeans,
    palette="viridis",
    s=70,
```

```
    legend="full"
)
plt.scatter(
    kmeans.cluster_centers_[:, 0],
    kmeans.cluster_centers_[:, 1],
    s=200,
    c="red",
    marker="X",
    label="Centroids"
)
plt.xlabel("Feature 1 (Standardized)")
plt.ylabel("Feature 2 (Standardized)")
plt.title("K-Means Clustering on Bank Marketing Dataset")
plt.legend(title="Cluster")
plt.tight_layout()
plt.show()
```



In [7]:
```
print("Cluster assignments for first 10 samples:")
for i in range(10):
    print(f"Sample {i + 1}: Cluster {y_kmeans[i]}")
```

```
Cluster assignments for first 10 samples:
Sample 1: Cluster 0
Sample 2: Cluster 0
Sample 3: Cluster 0
Sample 4: Cluster 0
Sample 5: Cluster 0
Sample 6: Cluster 0
Sample 7: Cluster 0
Sample 8: Cluster 0
Sample 9: Cluster 0
Sample 10: Cluster 0
```

In [ ]: